

Ceren Sahin

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EDUCATION

University of Toronto, St. George Campus

Sep 2023 – present

Bachelor of Science (BSc), Mathematics Specialist (Honors in the US), Computer Science Minor, expected 2027

Cumulative GPA: 3.89, Math GPA: 3.94, Math GPA percentage: 90.1 %

RESEARCH INTERESTS

Representation Theory, Algebraic Number Theory, p-adic Lie Theory

RESEARCH EXPERIENCE

Undergraduate Researcher - Analytic Number Theory

May 2026 - Aug 2026

Supervisor: Prof. Asif Zaman, University of Toronto

- Worked on finding a lower bound for the conductor of Dirichlet L-functions whose first zero has imaginary part less than 1, computing various test functions using the explicit formula that links the zeros of a Dirichlet L-function to a specific prime sum.
- Co-authored the [research manuscript](#), summarizing our computations and findings.

Undergraduate Researcher- Representation Theory

May 2025 - Dec 2025

Supervisor: Prof. Mathilde Gerbelli-Gauthier, University of Toronto

- Established a bijective correspondence between two different parametrisations of cohomological representations of the unitary group $U(p, q)$, developed independently by M. Gerbelli-Gauthier and N. Bergeron.
- Co-authored a [research manuscript](#) detailing the construction and proof of the bijection.
- Studied the first 14 chapters of *Fulton & Harris, Representation Theory: A First Course* during the first two months of the meetings.
- This summer project continued as a reading course on group cohomology during Fall 2025 academic season.

READING COURSES

Reading Course – Representations of $SL_n(\mathbb{Z})$ for $n > 3$

May 2026 - Aug 2026

Supervisor: Prof. Daniel Litt, University of Toronto

- Read first nine chapters of *Analytic Pro-p Groups*, Dixon et. al and typed detailed exercise solutions.
- Wrote an [expository paper](#) on the proof of superrigidity principle for $SL_n(\mathbb{Z})$, adapting the argument by Bass, Milnor and Serre to this special case.

Reading Course – Algebraic Combinatorics & Lie Algebras

Jan 2025 - Apr 2025

Supervisor: Prof. Hunter Spink, University of Toronto

- Read the first five chapters of *Algebraic Combinatorics* by Richard Stanley and the first eight chapters of *Introduction to Lie Algebras* by Karin Erdmann.
- Wrote comprehensive notes on finite dimensional representations of $\mathfrak{sl}(2, \mathbb{C})$ and explored vector fields on $\mathbb{R}^2$ generated by exponentiation and Möbius transformations.

GRADUATE COURSEWORK

CORE COURSES

- Real Analysis I (in progress)
- Algebra I (in progress)
- Algebraic Topology

ELECTIVE COURSES

- Algebraic Geometry (in progress)
- Algebraic Number Theory (in progress)
- Representation Theory
- Riemannian Geometry [[final paper](#) on the classification of locally symmetric Riemannian spaces]

AWARDS AND FUNDING

- University of Toronto Math Department Research Award - Undergraduate research funding 7,000 CAD (2026)
- University of Toronto Excellence Award — Undergraduate research funding, 7,000 CAD (2025)
- University of Toronto Scholars Award — Academic merit, 100,000 CAD (2023)

TEACHING EXPERIENCE

- MAT223 (Linear Algebra 1) Teaching Assistant. Led tutorials and graded assignments. Fall 2024 - present
- UT3-certified Student Tutor, providing tutoring for all first-year mathematics courses at UofT.